Datasheet

SKU: OMV-AE3-V1

Description

The OpenMV AE3 is a next-generation machine vision board built around the Alif Ensemble E3 system-on-chip, featuring dual ARM Cortex-M55 cores with Helium vector acceleration and a dedicated neural processing unit (NPU) delivering up to 250 GOPS of AI/ML compute. This enables real-time YOLO object detection at ~30 FPS entirely on-device. The board integrates a 1 MP global shutter camera, WiFi 802.11 a/b/g/n, Bluetooth 5.1, an 8x8 time-of-flight sensor, IMU, and microphone in a compact 8 g package. Programmable in MicroPython via the OpenMV IDE.

Target Areas

Edge AI, object detection, person counting, gesture recognition, industrial inspection, security, smart retail, autonomous systems, robotics.

Key Features

- **Alif Ensemble E3 SoC** — Dual ARM Cortex-M55 cores (400 MHz + 160 MHz) with Helium/MVE vector acceleration. Up to 3.2 GOPS per core.
- **On-device AI** — Dedicated NPU with 250 GOPS total (204 + 46 GOPS). Real-time YOLO object detection at ~30 FPS.
- **Global shutter camera** — 1 MP color global shutter sensor at up to 120 FPS. No rolling shutter distortion.
- **Integrated sensors** — 8x8 time-of-flight array (4 m range), accelerometer, gyroscope, and microphone built in.
- **WiFi + Bluetooth** — 802.11 a/b/g/n WiFi with chip antenna (U.FL alternative). Bluetooth 5.1 (BR/EDR and BLE).
- **USB-C High-Speed** — USB 2.0 High-Speed (480 Mbps) with EMI filtering and TVS protection.
- **Rich I/O** — 10 GPIO pins with SPI, I2C, I3C, CAN, 4x UART, and 2x 12-bit ADC. QWIIC connector.
- **Ultra-compact** — Only 8 g with four M1.6 mounting holes.

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1 Features and System Components

1.1 System-on-Chip

Component	Details
Alif Ensemble E3	Dual ARM Cortex-M55 cores with Helium/MVE vector acceleration. Core 0 runs at 400 MHz (3.2 GOPS, 8 ops/clock), Core 1 runs at 160 MHz (1.28 GOPS). Integrated NPU with 204 GOPS + 46 GOPS of AI/ML compute.

1.2 Memory

Component	Details
On-chip SRAM	13.5 MB SRAM.
On-chip MRAM	5.5 MB magnetoresistive RAM (non-volatile, byte-addressable).
Storage Flash	32 MB flash, 100 MHz 8-bit DDR mode (200 MB/s read bandwidth).

1.3 Connectivity

Component	Details
USB-C	USB 2.0 High-Speed (480 Mbps). EMI filtering and TVS protection on-board.
WiFi	802.11 a/b/g/n (up to 65 Mb/s). On-board chip antenna with U.FL connector as alternative.
Bluetooth	Bluetooth 5.1, BR/EDR and BLE.
SPI	SPI bus, up to 50 Mb/s.
I2C	2 I2C buses, up to 1 Mb/s each.
I3C	I3C bus, up to 10 Mb/s.
CAN	CAN bus, up to 8 Mb/s.
UART	4 asynchronous serial TX/RX interfaces.
QWIIC	Standard QWIIC connector for I2C peripherals. Compatible with Seeed Grove and Adafruit STEMMA.

1.4 Camera System

Component	Details
Image Sensor	1 MP color global shutter sensor. Up to 120 FPS.
Image Performance	~60 FPS at VGA (640x480) and below. Global shutter eliminates rolling shutter distortion for fast-moving objects.

1.5 Sensors

Component	Details
Time-of-Flight	8x8 ToF sensor array with 4-meter range for distance measurement and depth sensing.
IMU	Integrated accelerometer and gyroscope for motion sensing and orientation.
Microphone	Integrated on-board microphone for audio input.

1.6 User Interface

Component	Details
RGB LED	User-controllable and dimmable RGB LED.
User Button	1 user-programmable button.
Debug Switch	Recovery/debug mode switch.

1.7 Debug

Component	Details
SWD	6-pin SWD edge-connect header. Compatible with SEGGER J-Link debuggers.

1.8 Power Management

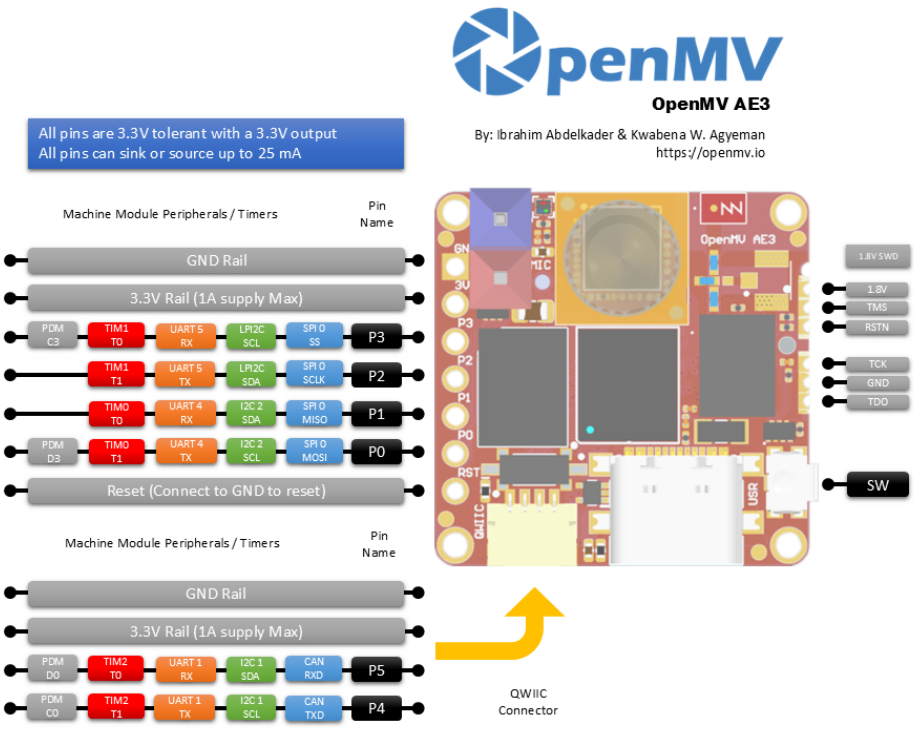
Component	Details
USB-C	Powered via USB-C connector.
RTC	On-board real-time clock with 4 KB backup RAM.

2 Ratings

2.1 Electrical Characteristics

Parameter	Description	Value	Unit
VIO	I/O voltage	3.3	V
VADC	ADC input voltage (max)	1.8	V
ADC	ADC resolution	12	bits
GPIO	GPIO count	10	pins

3 Connector Overview



B2B Header Pinout (on the board backside)

B2B	Name	Alternative Function						V Ref	Pin
Pin 2	P0	SPI 0 MOSI	I2C 2 SCL	UART 4 TX	TIM 0 T1	PDM D3		3.3V Ref	P5.1
Pin 4	P1	SPI 0 MISO	I2C 2 SDA	UART 4 RX	TIM 0 T0			3.3V Ref	P5.0
Pin 6	P2	SPI 0 SCLK	LPI2C SCA	UART 5 TX	TIM 1 T1			3.3V Ref	P5.3
Pin 8	P3	SPI 0 SS	LPI2C SCL	UART 5 RX	TIM 1 T0	PDM C3		3.3V Ref	P5.2
Pin 10	P4		I2C1 SCL	UART 1 TX	TIM 2 T1	PDM C0	CAN TXD	3.3V Ref	P0.5
Pin 12	P5		I2C1 SDA	UART 1 RX	TIM 2 T0	PDM D0	CAN RXD	3.3V Ref	P0.4
Pin 14	P6		I2C1 SDA	UART 3 CTS	TIM 9 T0			1.8V Ref	P7.2
Pin 16	P7		I2C1 SCL	UART 3 RTS	TIM 9 T1			1.8V Ref	P7.3
Pin 18	P8		I3C SDA	UART 3 RX	TIM 5 T0	ANA S10		1.8V Ref	P1.2
Pin 20	P9		I3C SCL	UART 3 TX	TIM 5 T1	ANA S11		1.8V Ref	P1.3
Pin 1	1.8V Supply								
Pin 3	P10	JTAG TCK						1.8V Ref	P4.4
Pin 5	P13	JTAG TMS						1.8V Ref	P4.5
Pin 7	P14	JTAG TDI						1.8V Ref	P4.6
Pin 9	P11	JTAG TDO						1.8V Ref	P4.7
Pin 11	JTAG RSTN							1.8V Ref	
Pin 13	RECOVERY							1.8V Ref	
Pin 15	3.3V Supply								
Pin 17	RSTN							3.3V Ref	
Pin 19	GND								

4 Functional Overview

4.1 Board Topology

Ref.	Description
U1	Alif Ensemble E3 SoC (dual Cortex-M55 + NPU)
CAM	1 MP global shutter camera module
ToF	8x8 time-of-flight sensor array
IMU	Accelerometer and gyroscope
MIC	On-board microphone
ANT	WiFi/Bluetooth chip antenna
USB	USB-C connector (High-Speed 480 Mbps)
QWIIC	QWIIC I2C connector
B2B	Board-to-board header
LED	Dimmable RGB LED

4.2 Power Tree

The OpenMV AE3 is powered via USB-C.

5 General Info

Dimension	Value
Weight	8 g
Mounting	Four M1.6 mounting holes
Operating Temperature	-40°C to +85°C

6 Certifications

Certification	Region	Details
RoHS	EU	Compliant with EU Directive 2011/65/EU (RoHS 2). All components and solder are lead-free.
REACH	EU	Compliant with EU Regulation (EC) No 1907/2006.

7 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

8 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/openmv-ae3
Documentation	https://docs.openmv.io
Community Forum	https://forums.openmv.io
Alif Ensemble E3	https://alifsemi.com/ensemble/

9 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.